THE UNIVERSITY OF TEXAS AT AUSTIN

Water Consumption Tracker

Industrial water monitoring, anomaly detection and remote data entry via Telegram.

Course	Python Programming
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CONTENTS

The Problem

Industrial facilities struggle to:

- Track consumption **across multiple areas**
- Detect **anomalies** and limit breaches in real time
- Log readings **without specialized software**
- Keep historical records **accessible and visual**

△

Manual entry errors, missed limit breaches, no mobile access.

Our Solution

A lightweight, browser-based tool:

- **Visual dashboards** for historical KPIs
- **Automatic anomaly detection** by area
- **Telegram bot** for remote data entry
- **CSV-backed storage** — no database needed

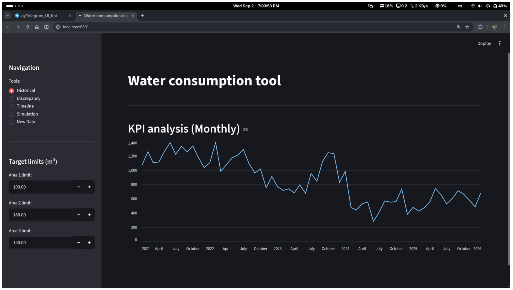

Fig. 1 — Dashboard overview: navigation panel and KPI chart.

Python · Streamlit · Telegram API

KPI Tracking

Monthly historical analysis

- Tracks consumption from **2021 to present**
- Automatic date parsing and sorting
- Single line chart — **easy to read trend**

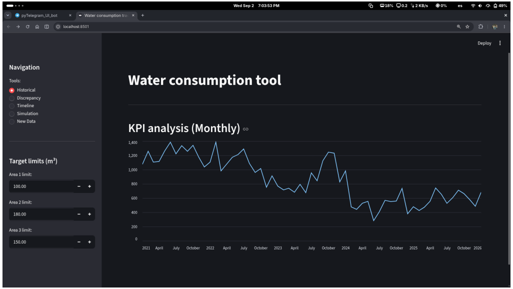

Fig. 2 — Monthly KPI chart: total consumption 2021–2026.

Anomaly Detection

Automatic limit breach detection

- Configurable **per-area thresholds** (m^3)
- Flags weeks with **negative Area 3** readings
- Anomalous weeks **highlighted in a table**
- Area 3 trend chart included for context

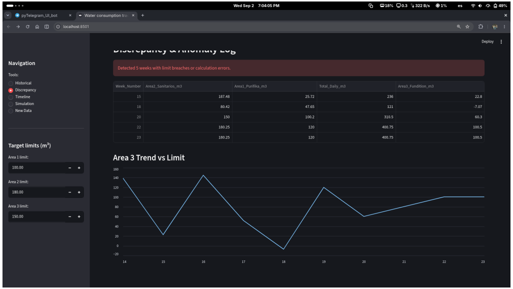

Fig. 3 — Discrepancy log: flagged weeks and Area 3 trend.

Shift Timeline

Day-by-day shift breakdown

- Compares **Shift 1, 2 & 3** simultaneously
- Interactive **timeframe slider** (days)
- **Pan** across any date window instantly
- Area chart reveals **peak consumption** patterns

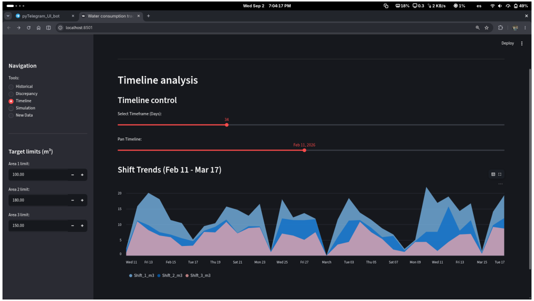

Fig. 4 — Timeline: shift trends Feb 11 – Mar 17, 2026.

Mobile Data Entry via Telegram

No app installation needed

- Send readings from **any phone**
- Two accepted formats:
  - `21, 280, 80.5, 90.5`
  - `sem 21 / total 280 / area1 80.5`
- Dashboard **fetches on demand**
- Data previewed, then **saved with one click**

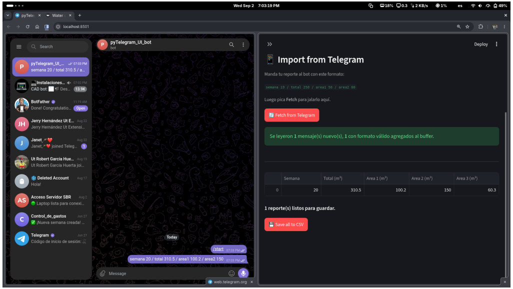

Fig. 5 — Telegram bot and import panel side by side.

What-If Simulation

Evaluate actions before implementing them

- Describe a proposed conservation action
- Input estimated **weekly savings** (m^3)
- Tool calculates **projected annual impact**
- Supports data-driven decisions with **no modeling required**

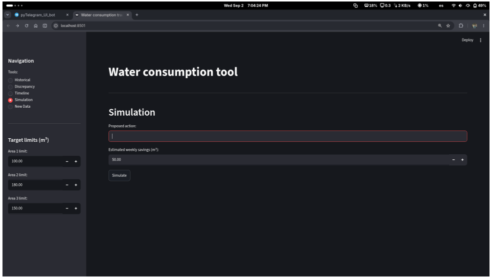

Fig. 6 — Simulation panel: action input and annual KPI projection.

Summary

Feature	Capability
Historical KPIs	Monthly trends since 2021
Anomaly detection	3 configurable area limits
Shift timeline	Day-level granularity
Remote entry	Telegram bot
Simulation	Annual impact forecast
Storage	CSV — portable, no setup

✓

Ready to use.

No installation,
no database,
no IT department.

Thank You

Questions?

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